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Sardar Patel University

SYBCA (3rd Sem) (NEP) Examination – OCT-2025

US03MABCA01|| Fundamentals of Data Structure

Date: 06/10/2025, Monday
Time: 02:00 pm to 04:30 pm **Total Marks:50**

Q.1 Multiple Choice Questions. **[10]**

1 __________of an array is the address of memory location where the first element of the array is located.
a. Size b. Type c. Base d. Index

2 _____________operation finds the presence of data items in the list of data items
a. Sorting b. Updating c. Merging d. Searching

3 Index of an array is always an ______value.
a. String b. Float c. Integer d. Stack

4 An operation that is used to give the value of an element at a particular position from a top of a stack is known as _____.
a. Push b. Pop c. Change d. Peep

5 The ______notation in which the operator is written before the operands.
a. Prefix b.. postfix c. infix d. None of above

6 The line formed for paying fees for college is an example of ______.
a. Stack b. simple queue c. circular queue d. priority queue

7 A binary tree has at most _____child.
a. One b. three c. two d. none of these

8 The nodes which have the same parent are called .
a. Degree. b Leaf. c. Height. d. Siblings.

9 Each node in singly linked list has____ fields.
a. 1 b. 3 c. 4 d. 2

10 ___ is the process to arrange the data in ascending or descending order.
a. Updating b. Searching c. Indexing d. Sorting

Q-2 Define array. Explain 1-D array in detail **[10]**

OR

Q-2 Explain in brief Classification of Data Structure and also write advantages and disadvantages of it. **[10]**

Q-3 Define Queue and write an algorithm to Insert and delete an element in a Simple queue. **[10]**

OR

Q-3 Define Stack. List applications of stack and write an algorithm for PUSH and POP operation of a stack. **[10]**

Q-4 Discuss the internal memory representation of a binary tree using sequential and linked representation. [10]

OR

Q-4 Write down an algorithm for inorder and postorder traversal of a binary tree. [10]

Q-5 What is searching? List searching techniques and explain ANY ONE of them with algorithm. [10]

OR

Q-5 Define Linked List. List and explain types of linked lists with example [10]

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